Why Basel Survives Heavy Tails:

The Hamerle–Rösch Self-Correction in Misspecified Credit Models

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Abstract

The Gaussian copula has zero tail dependence. Taken literally, that property should disqualify it as a description of joint default behaviour under stress, yet the Basel II IRB framework, which rests on the Gaussian Asymptotic Single Risk Factor model, has produced regulatory capital figures that have held up across two decades of testing including the 2007–2009 crisis. This paper traces the reconciliation to the calibration loop: when defaults follow a one-factor t-copula with finite degrees of freedom and one fits a Gaussian model by quasi-MLE, the asset correlation estimator absorbs the missing tail dependence by inflating, and the biased estimate delivers a loss-distribution forecast close to the true one. The result, due to Hamerle and Rösch (2005), is replicated here at the same publishable standard. The bias function is then charted on a finer grid in $(\rho_{\text{true}}, \nu_{\text{true}})$, loss-distribution forecasts are compared across three regimes, and four conditions under which the compensation fails or becomes inapplicable are identified. The empirical implications for the calibration of asset correlation on large mortgage panels are flagged but not resolved. The paper does not take a position on the adequacy of the regulatory framework. It isolates a specific mechanism and the empirical implications that would follow if that mechanism is present.

1 Introduction

The Asymptotic Single Risk Factor model behind the Basel II Internal Ratings-Based capital framework (Basel Committee on Banking Supervision, 2005; Gordy, 2003; Vasicek, 2002) assumes that the dependence structure between obligor defaults follows a Gaussian copula. Anyone who has worked with the model long enough knows this assumption is uncomfortable: the lower tail dependence coefficient of the bivariate Gaussian, $\lambda_L = \lim_{u \to 0^+} \mathbb{P}(U < u \mid V < u)$, equals zero (Embrechts et al., 2002; Nelsen, 2006). Two assets that are jointly Gaussian, no matter how strong their linear correlation, behave as independent in the deep tail. For credit, this means that the probability of two obligors defaulting together vanishes faster than it should when default rates get small. The t-copula does not have this defect: its lower tail dependence is strictly positive whenever the degrees-of-freedom parameter ν is finite, and grows as ν decreases (Frey et al., 2001).

If one reads this property literally, the prediction is brutal. A capital framework built on a copula that decouples extreme co-movements ought to underestimate joint default risk in stress periods. And it should fail exactly when capital matters most, since Basel II targets the 99.9% confidence level, deep into the tail where Gaussian and t copulas part ways. Any sign of moderate tail dependence in the data-generating process should leave the Gaussian-based capital figure short.

The data tell a different story. Basel II was finalised in June 2004 and rolled out across major jurisdictions starting in 2007, going live just as the global financial crisis arrived. US mortgage defaults reached levels not seen since the 1930s. Corporate default rates spiked across investment grade and high yield. The IRB capital framework for plain credit exposures, residential mortgage, retail, corporate, sovereign, held. Banks lost money, sometimes a great deal, but the systematic pillar-1 failures that post-crisis reviews identified were elsewhere: operational risk, the trading book, OTC counterparty exposures. The structured-credit collapse, where the Gaussian copula and the Li model produced spectacular damage, is its own story (Brigo et al., 2010); the same copula assumption underlies both, yet only one of them broke. That asymmetry is what needs explaining.

The explanation is in Hamerle and Rösch (2005). When defaults are generated by a t-copula with true parameters $(\rho_{\text{true}}, \nu_{\text{true}})$ and one fits a Gaussian model by quasi-MLE on the resulting default panel, the estimator $\hat{\rho}_G$ is biased upward, and the bias depends in a specific way on ν_{true}: heavier tails produce more inflation. The biased estimate, fed back into the Gaussian model, produces a loss-distribution forecast that stays close to the true t-copula across the body of the distribution and through the moderate quantiles. The Gaussian copula does not become a better model under calibration; the calibration loop wraps around it and absorbs part of the missing tail dependence. That is how the capital framework keeps producing workable numbers despite a parametric defect that, in isolation, should disqualify it.

This paper replicates the H–R simulation in its original form, extends it along a finer grid in $(\rho_{\text{true}}, \nu_{\text{true}})$ to chart the bias surface, and compares loss-distribution forecasts under three regimes: the true t-copula, the biased Gaussian, and the naïve Gaussian where one uses ρ_{true} as input. The compensation is real but partial. It is also fragile. Section 7 discusses four conditions under which the compensation fails or becomes inapplicable, namely panel structure, single-parameter representation under joint stress, vintage homogeneity, and falsifiability against data, each of which matters for the empirical use of the framework. Whether the H–R prediction holds on actual mortgage panels is an empirical question that the consistency map of Section 6 frames but does not settle.

2 Model and notation

The data-generating process is the one-factor t-copula in default-mode form. For each period $t = 1, \ldots, T$, the latent variable of obligor $i \in \{1, \ldots, N\}$ is

$$Y_{it} \;=\; \sqrt{\rho}\, Z_t \;+\; \sqrt{1-\rho}\, \varepsilon_{it}, \tag{1}$$

with $Z_t \sim \mathcal{N}(0,1)$ the systematic factor shared by all obligors at time t, $\varepsilon_{it} \sim \mathcal{N}(0,1)$ the idiosyncratic shock specific to each obligor, and Z_t and ε_{it} mutually independent. The asset correlation $\rho \in [0,1]$, the Pearson correlation between the latent variables, controls how much of the latent variation is systematic. The use of Pearson correlation in this context is not the usual choice for copula-based modelling, where rank-based measures such as Spearman's ρ_S or Kendall's τ_K are preferred for their invariance under monotone transformations of the marginals (Embrechts et al., 2002). The parameter ρ retains its conventional meaning here because the latent marginals are fixed by construction to be standard Gaussian, which removes the marginal-dependence ambiguity that motivates the rank-based alternatives elsewhere. The Basel IRB framework is parametrised in these terms, and the H–R simulation must be replicated within the same parametrisation.

To move from Gaussian to t-copula one adds a chi-squared mixing factor $W_t \sim \chi^2_\nu$, independent of (Z_t, ε_{it}), and scales the Gaussian latent variable accordingly:

$$T_{it} \;=\; \frac{Y_{it}}{\sqrt{W_t/\nu}} \;=\; \frac{\sqrt{\rho}\,Z_t \;+\; \sqrt{1-\rho}\,\varepsilon_{it}}{\sqrt{W_t/\nu}}. \tag{2}$$

The marginals of T_{it} are then t_ν and the joint distribution carries the t-copula structure. Default of obligor i at time t occurs when T_{it} falls below the calibrated threshold $c_t = t_\nu^{-1}(\lambda)$, where λ is the marginal default probability over the horizon. By construction $\mathbb{P}(T_{it} < c_t) = \lambda$. A property of this construction worth flagging early is that the t-copula never reduces to independence: even when $\rho = 0$, the shared mixing factor W_t continues to couple the obligors in any period t. Dependence in the t-copula flows through two channels, the asset correlation ρ and the common chi-squared shock W_t/ν, and only the first one vanishes when ρ is set to zero.

Conditional on the realised pair (Z_t, W_t), obligors are independent, and the conditional default probability is

$$p^{(t)}(z, w; \lambda, \rho, \nu) \;=\; \Phi\!\left(\frac{c_t\sqrt{w/\nu} - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\right). \tag{3}$$

The Gaussian copula is the limiting case $\nu \to \infty$: then $W_t/\nu \to 1$ almost surely, $c_t \to \Phi^{-1}(\lambda)$, and one recovers the standard Vasicek conditional probability (Vasicek, 2002),

$$p^{(g)}(z; \lambda, \rho) \;=\; \Phi\!\left(\frac{\Phi^{-1}(\lambda) - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\right). \tag{4}$$

The estimator under the misspecified Gaussian assumption maximises the panel log-likelihood

$$\ell_G(\lambda, \rho) \;=\; \sum_{t=1}^{T} \log \int_{-\infty}^{\infty} \binom{N}{d_t} p^{(g)}(z; \lambda, \rho)^{d_t} \left(1 - p^{(g)}(z; \lambda, \rho)\right)^{N-d_t} \phi(z)\, dz, \tag{5}$$

where d_t is the realised default count at time t and ϕ is the standard normal density. The integral over the systematic factor is computed by Gauss–Hermite quadrature, and the optimisation runs Nelder–Mead with multiple starting points to keep the optimiser from settling into a local maximum when the data come from a heavy-tailed copula.

This is the estimation approach of Hamerle and Rösch (2005), used here to keep results comparable with theirs. Other estimators are available, in particular method-of-moments procedures that recover the parameters directly from the sample moments of the default rate. Each route embeds different distributional assumptions and behaves differently in terms of computational cost and identification, and the choice between them has consequences for how cleanly one can separate ρ from ν. The paper stays with the QML benchmark and does not pursue that comparison further.

A parallel restriction is on the copula family itself. The t-copula has symmetric tail dependence, with $\lambda_L = \lambda_U > 0$ for $\nu < \infty$. Credit data may exhibit asymmetric joint behaviour, with co-movement concentrated in the lower tail, where joint defaults occur, and absent in the upper tail. Asymmetric alternatives such as the Clayton copula or its rotations can capture lower-tail dependence without imposing upper-tail dependence, and have been used elsewhere in credit applications. The paper retains the t-copula because the H–R benchmark and the Basel IRB calibration both sit within the elliptical family, and the goal here is to characterise the QML self-correction within that family; an extension to asymmetric copulas is a natural direction for future work.

3 Replication of the H–R simulation

The original paper reports estimation results on a $2 \times 3 \times 3$ grid of (λ, ν, ρ) values, with $T = 10$, $N = 1000$, and $M = 1000$ replications per cell. Table 1 reports the present replication at the same publishable standard, $M = 1000$, with Nelder–Mead multistart at every fit so that the heavy-tailed cases do not collapse into local maxima. The bias in any given cell is the difference between the reported average and the true ρ heading its column.

Table 1. Replication of Hamerle and Rösch (2005), Table 1. Each cell reports the average $\hat{\rho}_G$ across $M = 1000$ replications under the Gaussian QML estimator, when the data-generating process is the one-factor t-copula at parameters $(\rho_{\text{true}}, \nu_{\text{true}})$. Standard deviations of the estimator across replications are in parentheses.

λ	ν	$\rho_{\text{true}} = 0.00$	$\rho_{\text{true}} = 0.02$	$\rho_{\text{true}} = 0.10$
0.010	5	0.423 (0.162)	0.438 (0.162)	0.445 (0.175)
0.010	10	0.259 (0.131)	0.269 (0.123)	0.314 (0.145)
0.010	30	0.081 (0.048)	0.100 (0.060)	0.174 (0.099)
0.002	5	0.362 (0.308)	0.323 (0.300)	0.319 (0.305)
0.002	10	0.346 (0.259)	0.347 (0.252)	0.344 (0.271)
0.002	30	0.122 (0.118)	0.149 (0.136)	0.221 (0.191)

The replication recovers the H–R bias pattern at the published standard. For the $\lambda = 0.01$ row, the difference between the present numbers and the original is at most 0.024 across all nine cells, well within Monte Carlo error given the standard deviations reported under $M = 1000$. The pattern is monotonic in ν and rises with ρ_{true}, as the original paper documents.

Two features of the table deserve to be flagged. The first is monotonicity: heavier tails inflate $\hat{\rho}_G$, robustly across both default probability levels and across all three values of ρ_{true}. Reading down the first column at $\lambda = 0.01$, $\hat{\rho}_G$ moves from 0.081 at $\nu = 30$ to 0.259 at $\nu = 10$ to 0.423 at $\nu = 5$, against a true value of zero. The second is more subtle. When ν is low, the dependence on ρ_{true} almost disappears. At $\lambda = 0.01$ and $\nu = 5$, the three columns fall within a band of 0.022 around 0.435, regardless of whether the true asset correlation is zero, two percent, or ten percent. This is effective non-identification: under heavy-tailed data, the QML estimator informs more about the degrees of freedom than about the correlation parameter it nominally targets.

The standard deviation of $\hat{\rho}_G$ grows substantially at $\lambda = 0.002$. With low marginal default probabilities and $T = 10$, several panels in any given replication contain periods with zero defaults, and the likelihood becomes nearly flat in ρ for those samples. The estimator inherits that flatness as variance. This is a small-sample identification issue, not a feature of the H–R mechanism, and it is worth keeping in mind when reading the second row of the table.

4 The bias function

The replication grid above is sparse in ν. To see how the bias evolves across the range from heavy-tailed to near-Gaussian, the experiment is repeated on a finer grid,

$$\nu \in \{3, 4, 5, 6, 7, 8, 10, 12, 15, 20, 25, 30, 40, 50, 70, 100\},$$

at five values of $\rho_{\text{true}} \in \{0.000, 0.025, 0.050, 0.075, 0.100\}$, with $\lambda = 0.01$ and $M = 500$ replications per cell. Figure 1 plots the average $\hat{\rho}_G$ as a function of ν_{true}.

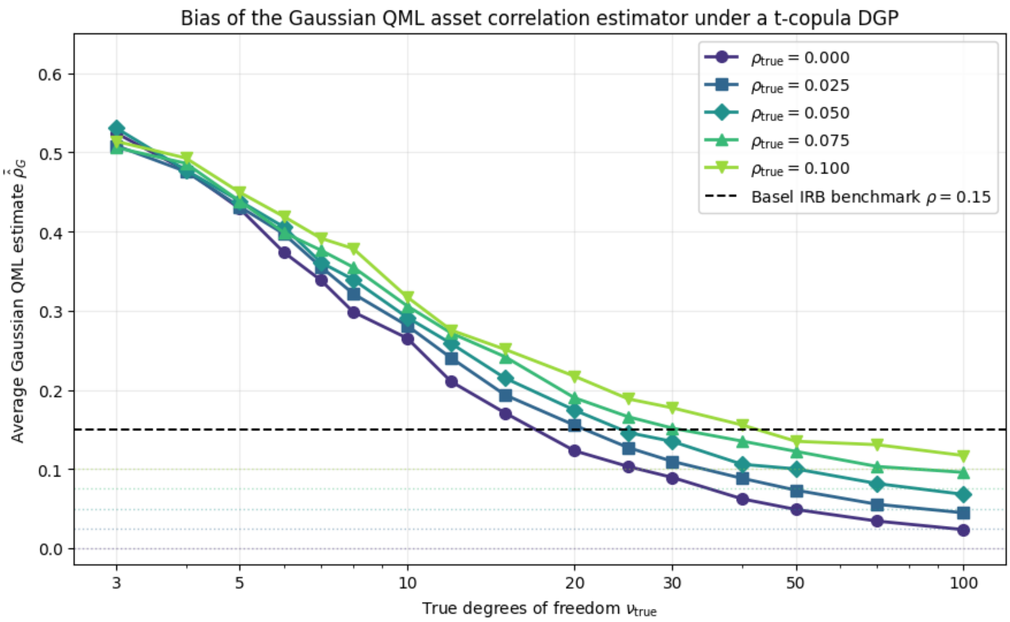

Figure 1. Bias of the Gaussian QML asset correlation estimator under a one-factor t-copula data-generating process. Each curve corresponds to a fixed value of ρ_{true}. The horizontal dashed line at $\rho = 0.15$ marks the Basel IRB benchmark for residential mortgages. Three regions are visible: a near-Gaussian regime ($\nu \gtrsim 30$) where the estimator is approximately unbiased, a transition regime ($\nu \in [10, 30]$) where the QML inflation crosses the Basel benchmark, and a heavy-tailed regime ($\nu \lesssim 10$) where the estimator becomes informative about ν_{true} rather than ρ_{true}.

Three regions show up in the figure, and each one has a clean mechanical reading.

The right-hand region: consistency under correct specification. For $\nu \geq 30$ the five curves separate cleanly and converge toward their respective ρ_{true} values. The curve for $\rho_{\text{true}} = 0$ approaches zero, the curve for $\rho_{\text{true}} = 0.10$ approaches the neighbourhood of 0.10, and the intermediate curves separate in between. This is the basic asymptotic story of QML: as $\nu \to \infty$ the t-copula converges to the Gaussian, and the QML estimator built on the Gaussian assumption becomes consistent for ρ_{true}. The lesson is operational. Misspecification matters only when the model is, in fact, misspecified. If one's prior is that defaults are well described by a near-Gaussian structure, the QML estimator delivers what it claims to deliver and the H–R bias is moot. The heavy-tailed regime to the left is what gives the entire mechanism its practical bite.

The transition region: where Basel sits. The curves cross the Basel IRB benchmark of $\rho = 0.15$ in the parameter region $\nu \in [15, 30]$. Calibrating the Gaussian model to data drawn from a process with ν in this range and reading off $\hat{\rho}_G$ produces, mechanically, a number close to 15%.

The interpretation needs care. The IRB framework was finalised in 2004, and the regulatory value of 0.15 for residential mortgage exposures predates the H–R paper by a year. The framework was calibrated with a mix of supervisory judgement, industry calibrations, and historical default data, not by reverse-engineering a t-copula DGP. The relevant statement is the converse. If historical ρ estimates from credit panels cluster near the Basel benchmark, that clustering is consistent with a world in which the true data-generating process has ν in the moderate range and ρ_{true} low to moderate. The Basel value is a fixed point of the calibration loop under that DGP, not because anyone designed it to be, but because that is what the QML estimator delivers.

The left-hand region: effective non-identification. For $\nu \leq 5$ the five curves are nearly indistinguishable. At $\nu = 3$, the average $\hat{\rho}_G$ falls in a tight band between roughly 0.51 and 0.53 across all five values of ρ_{true}, including the extremes at zero and ten percent. The practitioner's intuition runs as follows. When the true copula is very heavy-tailed, the realised default rate swings hard from year to year even when the underlying asset correlation is close to zero. The Gaussian model has no chi-squared mixing variable to produce those swings, so it has to find them somewhere inside its own machinery, and the only available parameter is the asset correlation. The QML procedure inflates $\hat{\rho}_G$ until the implied volatility of the conditional default rate matches what the data show. Two very different generating processes, one with $\rho_{\text{true}} = 0$ and another with $\rho_{\text{true}} = 0.10$, both at $\nu = 3$, yield similar realised default-rate volatility and therefore similar $\hat{\rho}_G$. The asset correlation parameter has, in this regime, lost its identifying content. It is no longer an estimate of the true correlation; it is a summary statistic of the realised volatility, filtered through a misspecified parametric form.

The three regions taken together give a usage rule for any empirical $\hat{\rho}_G$. At one extreme, the estimator identifies the true correlation. At the other, it identifies the inverse of the degrees of freedom. In between, the estimate is a confound that depends on both. Without independent information on ν, an empirical $\hat{\rho}_G$ cannot be read unambiguously as either.

5 Loss distribution comparison

The bias function shows that $\hat{\rho}_G$ is inflated, but it does not by itself tell us whether the inflated estimate produces an accurate forecast of portfolio loss. To address this, three loss distributions are compared on the same panel structure ($N = 1000$, $\lambda = 0.01$):

1. the true t-copula at $(\rho_{\text{true}} = 0.05, \nu_{\text{true}} = 10)$;
2. the misspecified Gaussian at the average $\hat{\rho}_G$ that the QML procedure delivers on data from regime (1), namely $\hat{\rho}_G = 0.292$;
3. the naïve Gaussian at $\rho = \rho_{\text{true}} = 0.05$, which is what one gets if one ignores the t-copula structure entirely.

The summary statistics are in Table 2 and the densities in Figure 2.

Table 2. Risk measures of the portfolio loss distribution under three regimes, evaluated at $N = 1000$, $\lambda = 0.01$, with 50,000 Monte Carlo scenarios.

Regime	Mean	VaR 99%	VaR 99.9%	ES 99%
True t-copula	0.99%	9.3%	17.4%	12.9%
Biased Gaussian	1.00%	10.2%	21.5%	15.2%
Naïve Gaussian	1.00%	3.4%	5.2%	4.1%

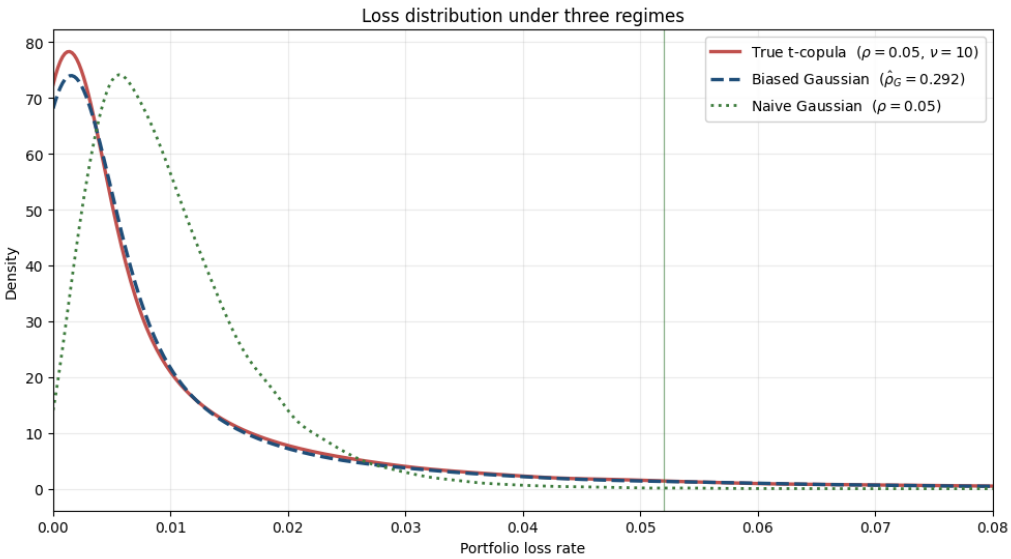

Figure 2. Loss distribution of the portfolio under three regimes. The true t-copula (solid) and the biased Gaussian (dashed) track each other through the body of the distribution; the naïve Gaussian (dotted) is visibly thinner. The vertical line marks the mean across regimes.

The two Gaussian regimes differ only in the value of ρ used as input. The difference between the biased Gaussian (dashed) and the naïve Gaussian (dotted) is therefore entirely the effect of the H–R bias. The biased version sits close to the true t-copula across the body of the distribution and through the 99% quantile. The naïve version is dramatically too thin, underpredicting loss at every quantile by a factor of two to three.

The compensation is partial. At VaR 99%, the biased Gaussian sits 0.9 percentage points above the true t-copula, conservative by roughly 10%. At VaR 99.9%, the gap widens to 4.1 percentage points, conservative by roughly 24%. At Expected Shortfall 99% the gap is 2.3 percentage points, conservative by roughly 18%. This is a structural pattern, not a numerical accident. No amount of upward bias in $\hat{\rho}_G$ can reproduce the asymptotic linkage that a t-copula imposes on extreme co-movements, because the Gaussian copula has $\lambda_L = 0$ regardless of ρ. For risk measures that probe deeper into the tail, the misspecification reasserts itself.

Stated plainly: when the true generating process has heavy tails, calibrating QML and using the inflated $\hat{\rho}_G$ produces a more accurate loss forecast than knowing ρ_{true} exactly and using the wrong copula. The compensation is asymmetric across regulatory frameworks. For Basel IRB capital at the 99.9% confidence level, the H–R compensation is a partial but material cushion. For Solvency II at 99.5% it works better, because the gap between biased Gaussian and true t-copula is smaller at that quantile. For stress testing at extreme quantiles, or for Expected Shortfall metrics, it offers progressively less protection.

6 Consistency map: where empirical estimates fall

The bias function in Section 4 fixed five values of ρ_{true} and varied ν_{true} along a single axis. The bias is in fact a function of two parameters, and reading it on a single axis loses information. Figure 3 shows the full bias surface as a contour map over the $(\rho_{\text{true}}, \nu_{\text{true}})$ plane, computed at $\lambda = 0.01$.

A few words on how to read the figure are in order, since each axis encodes a different concept.

The horizontal axis is ν_{true}, the degrees-of-freedom parameter of the t-copula generating the data. It is on a logarithmic scale running from 3 (very heavy tails) up to 100 (effectively Gaussian). The intermediate value $\nu = 10$ is the moderate-tail case used for the loss-distribution comparison in Section 5.

The vertical axis is ρ_{true}, the true asset correlation in the data-generating process. It runs from zero up to the Basel IRB benchmark of 0.15.

The colour scale and the contour lines both encode the same quantity: the average value of $\hat{\rho}_G$ that Gaussian QML would deliver when applied to data generated at the corresponding $(\rho_{\text{true}}, \nu_{\text{true}})$ pair. Cool colours correspond to low $\hat{\rho}_G$, warm colours to high $\hat{\rho}_G$. The black contour lines connect points that produce the same $\hat{\rho}_G$, and they are labelled from $\hat{\rho}_G = 0.05$ in the lower-right to $\hat{\rho}_G = 0.50$ in the upper-left.

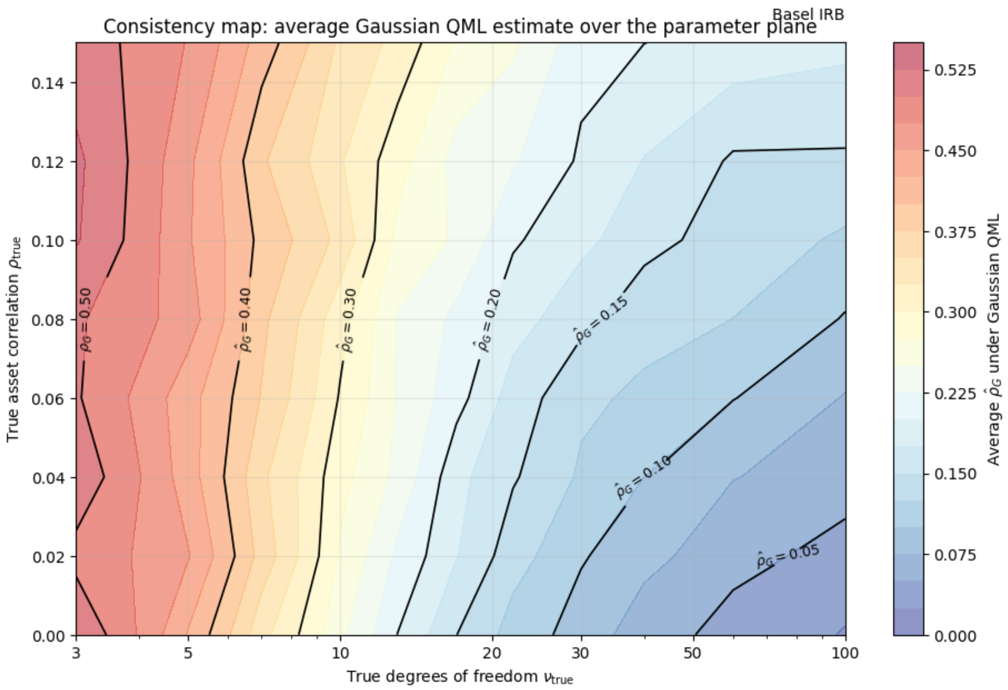

Figure 3. Average Gaussian QML estimate $\hat{\rho}_G$ as a function of the true data-generating parameters $(\rho_{\text{true}}, \nu_{\text{true}})$. Each contour identifies the set of true parameters consistent with a hypothetical empirical observation. The horizontal dashed line at $\rho_{\text{true}} = 0.15$ marks the Basel IRB benchmark.

The figure answers a forward question: given a world with true parameters $(\rho_{\text{true}}, \nu_{\text{true}})$, what does my Gaussian calibration report? Three examples make the mechanism concrete.

Take first a world with $\rho_{\text{true}} = 0.10$ and $\nu_{\text{true}} = 50$, in the upper-right corner: moderate correlation, mild tail dependence. The point lies in the light-blue region between the $\hat{\rho}_G = 0.10$ and $\hat{\rho}_G = 0.15$ contours. The Gaussian calibration on data from this world reports a number close to the true 10%; the bias is small because the underlying copula is close to Gaussian.

Take next a world with $\rho_{\text{true}} = 0.05$ and $\nu_{\text{true}} = 5$, in the lower-left: low correlation, very heavy tails. The point lies in the orange-red region near the $\hat{\rho}_G = 0.40$ contour. The Gaussian calibration reports around 40%, eight times the true correlation. The estimator has absorbed the tail dependence by inflating $\hat{\rho}_G$ massively.

Take finally a world with $\rho_{\text{true}} = 0.02$ and $\nu_{\text{true}} = 100$, in the lower-right: low correlation, near-

Gaussian dependence. The point is in the dark-blue region near the $\hat{\rho}_G = 0.05$ contour. The calibration reports approximately the truth, because there is no tail dependence to compensate for and the correlation is small to begin with.

The reverse reading is the one that matters in practice. A practitioner does not observe $(\rho_{\text{true}}, \nu_{\text{true}})$; those are parameters of nature. The practitioner observes $\hat{\rho}_G$, the output of a calibration. To use the figure in this direction, one fixes a value of $\hat{\rho}_G$, say 0.15 (the Basel benchmark), and traces the corresponding contour across the plane. That contour is a curve, not a point. It picks out an entire one-dimensional family of $(\rho_{\text{true}}, \nu_{\text{true}})$ pairs that all produce the same observed $\hat{\rho}_G$.

Reading the $\hat{\rho}_G = 0.15$ contour from upper-left to lower-right, an observation at the Basel benchmark is compatible with:

- a world with $\rho_{\text{true}} \approx 0.10$ and $\nu_{\text{true}} \approx 50$ (moderate correlation, mild tail dependence);
- a world with $\rho_{\text{true}} \approx 0.05$ and $\nu_{\text{true}} \approx 25$ (low correlation, moderate tail dependence);
- a world with $\rho_{\text{true}} \approx 0.00$ and $\nu_{\text{true}} \approx 17$ (zero correlation, pronounced tail dependence).

All three worlds produce $\hat{\rho}_G = 0.15$. The estimate alone does not tell which is the actual generating process. This is the under-identification problem made visible. The map from data-generating parameters to QML estimates is many-to-one, and inverting it needs more information than the estimate itself can give.

The three worlds have very different economic content. In the first, risk is dominated by the systematic factor and the Gaussian model is approximately correctly specified; capital based on $\hat{\rho}_G = 0.15$ is roughly accurate. In the third, risk is dominated by tail dependence and the true correlation is zero; the entire $\hat{\rho}_G = 0.15$ is bias, and capital built on it rests on the wrong foundation, with consequences for any risk measure that probes deep into the tail. The second world is intermediate. The same calibrated $\hat{\rho}_G$ can therefore correspond to materially different underlying risk structures, and its prudential meaning depends on which world is the data-generating one.

The map is therefore better read as a diagnostic that flags an under-identification problem than as a tool to solve it. Telling apart the $(\rho_{\text{true}}, \nu_{\text{true}})$ pairs compatible with a given observation requires information beyond the pooled QML estimate. Direct estimation of ν from the same panel, using the higher moments of the realised default rate, would in principle pin down the degrees of freedom and let one read off the corresponding ρ_{true} from the appropriate contour. Dynamic evidence on the stability of $\hat{\rho}_G$ across cyclical episodes would inform whether a single time-invariant correlation is empirically tenable. A decomposition of pooled estimates into within-cohort and between-cohort components would speak to whether the apparent correlation comes from the systematic factor of the model or from drift in portfolio composition. Each is an empirical question of its own, and each requires data of a quality and length that not every credit panel provides.

What this paper does not claim is that any of these identification routes will deliver clean answers in practice. Whether the higher-moment route is identified given typical sample sizes, whether dynamic stability holds across the cycles available in the historical record, whether vintage decomposition can be done on portfolios where origination cohorts overlap with macroeconomic regimes, all of these are open questions. The map flags the problem without resolving it.

7 Where the compensation fails

One clarification matters before going further. The H–R framework explains why Gaussian calibration produces workable numbers in the aggregate. It does not validate the Gaussian copula as a description of joint default behaviour. The mechanism is mechanical compensation between a misspecified parametric form and a biased estimator, and it is fragile in four directions. Each of them matters when the framework is put to empirical use.

Sensitivity to panel structure. The compensation depends on T and N. The H–R simulation holds these fixed at $T = 10$, $N = 1000$. Real portfolios depart from this in both directions. Long histories with deep recessions sample the lower tail more aggressively than the simulation implies, while highly granular portfolios reduce the binomial sampling variance and let small differences between $p^{(g)}$ and $p^{(t)}$ surface as identifiable bias signal. The strength of the compensation documented in Section 5 is therefore not a property of the framework as such; it depends on the panel on which the framework is applied, and it has to be assessed empirically for any specific data structure.

Sensitivity to single-parameter representation. The compensation reported in Section 5 operates within a fixed parameter pair $(\rho_{\text{true}}, \nu_{\text{true}})$. The QML procedure delivers one $\hat{\rho}_G$ that summarises the entire panel. In practice, real credit episodes contain stretches under which a single-parameter summary is informationally inadequate, regardless of whether the parameter is biased or unbiased. The 2007–2009 crisis illustrates this in adjacent markets. As Brigo et al. (2010) document in their post-crisis forensic study of structured-credit pricing models, implied asset correlations in CDO markets shifted dramatically across the dislocations of that period, moving from approximately 20% pre-crisis to above 70% at the peak. Whatever the right interpretation of those shifts, whether regime change, drift in the underlying collateral, breakdown of homogeneity assumptions on which any single-correlation summary depends, or some combination of these, the data did not lend themselves to summary by a single calibrated $\hat{\rho}_G$. The H–R compensation is a property of stable parametric environments. It does not address, on its own, the broader question of whether single-parameter representations are the right tool when the conditions of stability stop holding.

Sensitivity to vintage homogeneity. This is arguably the most important of the four fragilities for empirical work, and the one least appreciated in the practitioner literature. The H–R compensation assumes vintage homogeneity within the calibration panel: the parameters $(\rho_{\text{true}}, \nu_{\text{true}})$ are taken to be constants that apply uniformly to every obligor in every period. This assumption is rarely defensible for real credit portfolios. Origination standards, underwriting practices, regulatory regimes, and macroeconomic conditions at origination all vary across cohorts, and any of these can produce a true correlation that differs across vintages within a single pooled panel.

When that happens, an inflated $\hat{\rho}_G$ admits a competing reading that has nothing to do with tail dependence. Heterogeneity across cohorts produces dispersion in realised default rates that the QML estimator cannot tell apart from the dispersion produced by a heavy-tailed copula. Both inflate the apparent correlation. The two mechanisms, the H–R heavy-tailed compensation and pure vintage drift, generate observationally similar pooled estimates and very different policy conclusions. Under the H–R reading, an inflated $\hat{\rho}_G$ is a feature of the calibration loop and a partial cushion against tail risk. Under the vintage-drift reading, the same inflated $\hat{\rho}_G$ is an artefact of pooling heterogeneous cohorts, it bears no relationship to tail dependence, and provides no cushion of any kind.

Telling these apart requires within-cohort and between-cohort decomposition of the panel, which is an exercise the H–R framework does not perform and which any honest reading of an estimated $\hat{\rho}_G$ ought to do. The point is structural. A single pooled $\hat{\rho}_G$ is silent on which mechanism produced it. This is the fragility that most directly threatens the practical use of the H–R compensation as a defence of the regulatory framework. The compensation exists, but an observed $\hat{\rho}_G$ by itself cannot establish that this is the mechanism producing it.

Falsifiability. The H–R prediction is empirically testable. If the mechanism operates as described and dominates other sources of bias, large mortgage panels should produce $\hat{\rho}_G$ at or above the regulatory benchmark, with the size of the inflation reflecting the true degrees of freedom. The framework is not pre-crisis sophistication that the data later invalidated; it is a falsifiable prediction. Whether it holds on representative mortgage portfolios is an open empirical question, and the answer depends on how carefully one handles the three fragilities above.

8 Conclusion

Hamerle and Rösch (2005) document a mechanical compensation in the calibration loop wrapped around the Gaussian Asymptotic Single Risk Factor model. When the data-generating process has finite-degree heavy tails, the Gaussian QML estimator of asset correlation is biased upward, the bias depends systematically on the true degrees of freedom, and the resulting biased Gaussian forecast of portfolio loss tracks the true t-copula closely through the body of the distribution and the moderate-quantile region. The Basel IRB benchmark of $\rho = 0.15$ is reproduced mechanically by the QML self-correction in the parameter region $\nu \in [15, 30]$ with low-to-moderate true asset correlation. The mechanism is not part of the regulatory design. It arises because the Gaussian calibration absorbs part of the misspecification into the estimated correlation.

The compensation is partial, asymmetric across regulatory frameworks, and fragile in four directions: sensitivity to panel structure, sensitivity to single-parameter representation under joint stress, sensitivity to vintage homogeneity, and falsifiability against empirical data. These fragilities matter because the empirical use of the framework requires that the conditions under which the compensation operates be stated honestly.

Two extensions deserve mention. First, the elliptical-family restriction in this paper, both the Gaussian baseline and the t-copula alternative, imposes symmetric tail dependence by construction. Asymmetric copulas such as Clayton allow lower-tail dependence without upper-tail dependence and may be the more natural specification for default behaviour, where co-movement is concentrated in the loss region. Second, a convex combination of a Gaussian copula and a copula with tail dependence, $C_\alpha = (1-\alpha)\, C_{\text{Gauss}} + \alpha\, C_{\text{tail}}$, would retain the Gaussian's analytical convenience for the body of the distribution while tuning positive tail dependence through the mixture weight α. The lower tail dependence of such a construction is the convex combination of the components, $\lambda_L^{C_\alpha} = \alpha\, \lambda_L^{C_{\text{tail}}}$. Whether either of these directions delivers cleaner identification than the QML procedure under a single t-copula is an open question and a natural target for follow-on work.

The empirical question, whether data from large mortgage panels yield estimates that line up with the H–R prediction or fall instead in regions of the consistency map that require alternative explanations, is left open. The framing adopted here is one of exposition. The mechanism is documented, its scope is identified, its falsifiable consequences are flagged. Whether the empirical record bears them out is a different question, and one this paper does not attempt to settle.

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